

The pair singular series of a polynomial V : the dilated pieces below $H^{1/2}$

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Abstract

For a monic irreducible quadratic f without fixed prime divisor, part III decomposed the Cesàro off-diagonal $\text{Off}_f^*(H)$ into pieces indexed by the dilation u and proved that the pieces with $u > H^{1/2+\varepsilon}$ contribute a main term $c_{\text{off}}(f)H$. This paper treats dilations below that threshold. We prove that the pieces with $H^{0.47} < u \leq H^{0.53}$ contribute $O(H^{1-\delta})$, so that $\sum_{u > H^{0.47}} w(u) \mathcal{P}_u(H/u) = c_{\text{off}}(f)H + O(H^{1-\delta})$; the input is a bilinear estimate for Kloosterman fractions of Bourgain and Garaev together with a bound for the additive energy of reciprocals, valid for every modulus, which we deduce from an argument of Cilleruelo and Garaev. For negative discriminants we prove a second result at the other end: the pieces with $u \leq H^{1/3-c}$ contribute $O(H^{1-\delta(c)})$, by running the automorphic kernel method of Grimmelt and Merikoski with the leading coefficient u^2 tracked. What remains is a neighbourhood of $u = H^{1/3}$.

1 Introduction

The problem. Let $f = t^2 + bt + c \in \mathbb{Z}[t]$ be monic, irreducible, without fixed prime divisor, of discriminant D . Part III [6] wrote the Cesàro off-diagonal of the pair singular series as a sum of *pieces*,

$$\text{Off}_f^*(H) = P(1) \sum_u \lambda(u) \omega(u) \mathcal{P}_u(H/u),$$

one for each squarefree *dilation* u , the piece $\mathcal{P}_u(Y)$ being the same problem at length $Y = H/u$ with the roots of $x^2 \equiv D$ replaced by those of $u^2 x^2 \equiv D$. Each piece is $O(Y)$, and the weights satisfy $\sum_{u \leq H} |w(u)|/u \asymp \log H$: the trivial bound for the whole sum is $H \log H$, which is exactly the size of the statement to be proved. Every piece must therefore be small, and the difficulty is uniformity in u .

Exponent coordinates. Write $u = H^\alpha$. Each method available for the pieces gives a saving $H^{-s(\alpha)}$ over the trivial bound on an interval of α , and degenerates at the ends of that interval. In these coordinates part III proves $s > 0$ for $\alpha > 2/3$ (averaging the dilations along u), and the present author's extension of that theorem, recorded in the revised part III, gives $s > 0$ for $\alpha > 1/2$ by a dispersion argument with Weil's bound. Below $1/2$ both mechanisms fail: the incomplete Kloosterman sums that dispersion produces have modulus $[d_1, d_2] \asymp Y^2$, and Weil's bound is trivial once the range of u is shorter than the square root of that modulus. This paper supplies two further mechanisms, one on each side of $\alpha = 1/3$.

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Results. Theorem 5 treats α near $1/2$: the weight w is decomposed as $1 * \chi_D * \gamma$, and the resulting bilinear forms in u are estimated by Weil's bound when one factor is long (Type I) or short (Type II), and, in the remaining balanced case, by the bilinear bound of Bourgain and Garaev [1] for Kloosterman fractions. That bound needs an estimate for the additive energy of reciprocals; the one in [1] is too weak at the sizes that occur here, and Lemma 2 supplies a stronger one, valid for every modulus, by transferring an argument of Cilleruelo and Garaev [2] for points on modular hyperbolas. Theorem 7 treats small α for negative discriminants: piece u is, after removing the content, a Type I sum for the polynomial $u^2\ell^2 + |D|$ in the sense of Grimmelt and Merikoski [3], and their automorphic kernel theorem [4] applies at level u^2e once the two kernel quantities are estimated with the dilation tracked. The diagonal is the number of orbits of the family, $\asymp u$; the off-diagonal, which their general bound would give as $\asymp u^2$, is in fact bounded independently of u (Lemma 6), because the distances inside the family are quantised in steps of $1/(4h)$ whatever the dilation.

What remains. Both mechanisms degenerate at $\alpha = 1/3$, from opposite sides and for unrelated reasons: there the balanced Type II atoms sit at $d^{1/4}$, where the energy input is not strong enough, and the kernel method's diagonal $\asymp u$ meets the trivial bound. The final section explains what the residual band costs. Two points are worth recording. A band of width 2δ in α costs $\delta \cdot H(\log H)^{1-c} \log \log H$, not $\delta H \log H$, because part III's Theorem E applies piece by piece inside it; so a failing method never makes the total worse. And if the savings on both sides vanish only at $\alpha = 1/3$, with implied constants that are powers of $\log H$, then the blocks near the point contribute $O(H)$ in total, which would give $\text{Off}_f^*(H) = O(H)$ — Hypothesis (E) of part I [5], a bounded second-order term in the Cesàro asymptotic. The constant A_f needs strictly more: a saving at the point itself.

2 Notation

We follow part III [6]. For a prime p let $\omega(p)$ be the number of roots of f modulo p , $\lambda(p) = p/(p - 2\omega(p))$, $P_p = 1 - \omega(p)^2/(p - \omega(p))^2$, $P(1) = \prod_p P_p$ and $w(u) = P(1)\lambda(u)\omega(u)$; w is supported on squarefree u and $\sum_{u \leq t} |w(u)| \ll t$. A modulus d' is *admissible* if it is squarefree, all its prime factors are split, and $(d', 2Du) = 1$; $R_{d'}^{(u)} = \{x \bmod d' : u^2x^2 \equiv D\}$, $\rho_k(d'; a) = \sum_{r \in R_{d'}} e(kar/d')$, and $\langle x \rangle_{d'} \in [1, d']$ is the representative of x . With $B_{d'}^{(Y)}(m) = \sum_{h \leq Y, h \equiv m \pmod{d'}} (Y - h) - Y^2/(2d')$, the piece is

$$\mathcal{P}_u(Y) = \sum_{d' > 1} \sum_{\text{admissible}} \frac{\lambda(d')}{d'} \sum_{x \in R_{d'}^{(u)}} B_{d'}^{(Y)}(\langle x \rangle_{d'}), \quad Y = H/u.$$

We write $G_{d'}(a, Y) = \sum_{r \in R_{d'}} B_{d'}^{(Y)}(\langle ar \rangle_{d'})$, $\bar{G}_{d'}(Y)$ for its mean over the units a , $G^\circ = G - \bar{G}$, and $\xi_k(d', N) = (1 - \cos(2\pi kN/d'))/\sin^2(\pi k/d')$ for the coefficients of the finite Fourier expansion of $B^{(N)}$ (part III, Lemma finfourier). Throughout, x^* denotes the inverse of x modulo the modulus at hand, $e_m(z) = e(z/m)$, and $\varepsilon > 0$ is arbitrarily small. Parameters: $\delta = 10^{-3}$, $\eta = \delta/10$, $\delta_3 = \delta$, $\varepsilon' = \eta/7$, $\kappa = \delta/2$, $K_1 = H^{2\eta}$.

3 The energy of reciprocals and bilinear Kloosterman fractions

We write x^* for the inverse of x modulo the modulus at hand and $e_m(z) = e(z/m)$.

Lemma 1 (smooth incomplete Kloosterman sums). *Let $q \geq 1$ be squarefree, $a \in \mathbb{Z}$, $X > 0$ and $T \geq 1$, and let W be smooth, supported in an interval of length at most $4X$, with $\|W\|_\infty \ll 1$*

and $\int |W''| \ll T^2/X$. Then

$$\sum_{(v,q)=1} W(v) e_q(av^*) \ll \tau(q) \left(1 + \frac{X}{q}\right) T^2 q^{1/2} (a, q)^{1/2}.$$

Proof. Split v into classes modulo q and apply Poisson summation in each: the sum equals $\sum_h \widehat{W}(h/q) S(a, h; q)/q$ with $S(a, h; q)$ the Kloosterman sum. For squarefree q , Weil's bound at the primes and twisted multiplicativity give $|S(a, h; q)| \leq \tau(q) (a, h, q)^{1/2} q^{1/2}$, which covers $h = 0$, where $S(a, 0; q) = c_q(a)$ is a Ramanujan sum and $|c_q(a)| \leq (a, q) \leq q^{1/2} (a, q)^{1/2}$. Since $\widehat{W}(\xi) \ll \min(X, T^2/(X\xi^2))$, one has $q^{-1} \sum_h |\widehat{W}(h/q)| \ll T^2(1 + X/q)$. $\square$

Lemma 2 (additive energy of reciprocals, every squarefree modulus). *Let $m \geq 2$ be squarefree, $1 \leq N \leq m$, and $J_4(N) = \#\{x \in [1, N]^4 : (x_i, m) = 1, x_1^* + x_2^* \equiv x_3^* + x_4^* \pmod{m}\}$. Then for every $\varepsilon > 0$,*

$$J_4(N) \ll_\varepsilon m^\varepsilon (N^2 + N^{7/2} m^{-1/2}).$$

Proof. Step 1: a hyperbola in a square box. Let $(\lambda, m) = 1$, $K \in \mathbb{Z}$ and $1 \leq M \leq m$; we claim $I := \#\{(x, y) \in (K, K+M]^2 : xy \equiv \lambda \pmod{m}\} \ll_\varepsilon m^\varepsilon (M^{3/2} m^{-1/2} + 1)$. For $M = 1$ this is trivial, so let $M \geq 2$. Writing $x = K + x'$, $y = K + y'$ with $1 \leq x', y' \leq M$, the congruence becomes $x'y' + Kx' + Ky' \equiv b := \lambda - K^2 \pmod{m}$. Put $T = \lfloor (m/M)^{1/2} \rfloor \geq 1$; by Dirichlet's theorem there are $1 \leq t \leq T$ and u_0 with $tK \equiv u_0 \pmod{m}$ and $|u_0| \leq \min(m/T, m/2)$. Multiplying by t ,

$$tx'y' + u_0x' + u_0y' = b_0 + zm, \quad |b_0| < m/2, \quad |z| \leq \frac{TM^2}{m} + \frac{2M}{T} + \frac{1}{2} \leq 5M^{3/2} m^{-1/2} + \frac{1}{2}.$$

For each z this is $(tx' + u_0)(ty' + u_0) = n_z := t(b_0 + zm) + u_0^2$, and $|n_z| \leq 7m^2$. Now $n_z \equiv t^2(x' + K)(y' + K) \equiv t^2\lambda \pmod{m}$, and λ is a unit with $1 \leq t^2 \leq m/M < m$, so $m \nmid n_z$ and in particular $n_z \neq 0$. Each z therefore contributes at most $\tau(|n_z|) \ll_\varepsilon m^\varepsilon$ pairs, and summing over the $\ll M^{3/2} m^{-1/2} + 1$ admissible z gives the claim.

Step 2: from energy to hyperbolas. Let $J_2(\lambda) = \#\{x, y \leq N, (xy, m) = 1 : x^* + y^* \equiv \lambda\}$, so $J_4 = \sum_\lambda J_2(\lambda)^2$ and $\sum_\lambda J_2(\lambda) = N'^2 \leq N^2$ with $N' = \#\{x \leq N : (x, m) = 1\}$. For a unit λ the congruence $x^* + y^* \equiv \lambda$ is equivalent to $(x - \lambda^*)(y - \lambda^*) \equiv \lambda^{*2}$, a unit hyperbola in a square box with equal shifts, so Step 1 gives $J_2(\lambda) \ll m^\varepsilon (N^{3/2} m^{-1/2} + 1)$. If $g = \gcd(\lambda, m) > 1$ then $p \mid g$ forces $x \equiv -y \pmod{p}$, hence $x \equiv -y \pmod{g}$ and $g \leq 2N$; modulo m/g the residue λ is a unit, so Step 1 applied there gives $J_2(\lambda) \ll m^\varepsilon (g^{1/2} N^{3/2} m^{-1/2} + 1)$ when $N \leq m/g$, while $J_2(\lambda) \leq N$ always. Using $\sum_{\gcd(\lambda, m)=g} J_2(\lambda) \leq N(2N/g + 1)$ and summing over the divisors $g \leq 2N$ of m yields the lemma; in the range $N > m/g$ one uses $4N^4/m \leq 4N^{7/2} m^{-1/2}$. $\square$

Lemma 3 (Bourgain–Garaev). *Let $m \geq 2$, $(a, m) = 1$, $1 \leq N_1, N_2 \leq m$, and let α_1, α_2 be coefficients bounded by 1 and supported on units. Then for all $k_1, k_2 \geq 1$,*

$$\left| \sum_{x_1 \leq N_1} \sum_{x_2 \leq N_2} \alpha_1(x_1) \alpha_2(x_2) e_m(ax_1^* x_2^*) \right|^{2k_1 k_2} \leq m N_1^{2k_1 k_2 - 2k_1} N_2^{2k_1 k_2 - 2k_2} J_{2k_1}(N_1) J_{2k_2}(N_2).$$

Proof. This is the inequality established in the proof of Theorem 3 of [1]: two applications of Hölder's inequality followed by Cauchy–Schwarz, using $\sum_\lambda J_k(\lambda) \leq N^k$ and that a is a unit. $\square$

Corollary 4. *For squarefree m and $m^{1/3} \leq N_1, N_2 \leq m$, under the hypotheses of Lemma 3, $|\sum \sum \alpha_1 \alpha_2 e_m(ax_1^* x_2^*)| \ll_\varepsilon m^\varepsilon (N_1 N_2)^{15/16}$.*

Proof. Take $k_1 = k_2 = 2$ in Lemma 3 and insert Lemma 2: for $N_i \geq m^{1/3}$ the second term of the energy bound dominates, $N_i^{7/2} m^{-1/2} \geq N_i^2$, so the right-hand side is $\ll m^{1+4\varepsilon} (N_1 N_2)^4 (N_1 N_2)^{7/2} m^{-1} = m^{4\varepsilon} (N_1 N_2)^{15/2}$. $\square$

4 The pieces around $u = H^{1/2}$

Theorem 5. *Let f be as above. There is $\delta_M > 0$ (any $\delta_M < 4\eta/7$ with the parameters above, so $\delta_M = 5 \cdot 10^{-5}$ is admissible) such that*

$$\sum_{H^{0.47} < u \leq H^{0.53}} w(u) \mathcal{P}_u(H/u) \ll_f H^{1-\delta_M}.$$

Consequently, with part III's Theorem on the range $u > H^{1/2+\varepsilon}$ applied at $\varepsilon = 0.03$, $\sum_{u > H^{0.47}} w(u) \mathcal{P}_u(H/u) = c_{\text{off}}(f)H + O(H^{1-\delta_M})$.

Proof of Theorem 5. The proof occupies the rest of this section. It is organised as a reduction to a Mellin-separated bilinear form (Step R1), a decomposition of the weight w with a covering of the resulting boxes (Step R5), and three estimates: a Type I bound (Step R2), a Type II bound (Step R3) and the residual balanced case (Step R4), the last being where Corollary 4 is used.

Notation for the proof. Put $\delta = 10^{-3}$, $\eta = \delta/10$, $\delta_3 = \delta$, $\varepsilon' = \eta/7$, $\kappa = \delta/2$ and $K_1 = H^{2\eta}$; all implied constants depend on f and on these parameters. For an admissible modulus d' and a unit a let $G_{d'}(a, Y) = \sum_{r \in R_{d'}} B_{d'}^{(Y)}(\langle ar \rangle_{d'})$, let $\bar{G}_{d'}(Y)$ be its mean over the units and $G^\circ = G - \bar{G}$; let $\rho_k(d'; a) = \sum_{r \in R_{d'}} e(kar/d')$, $\epsilon_{d'}(k) = c_{d'}(k)/\varphi(d')$ and $\xi_k(d', N) = (1 - \cos(2\pi kN/d'))/\sin^2(\pi k/d')$, and set $b_k(d') = \min(1/(4k^2), \pi^2(2Y)^2/(4d'^2))$.

Step R1: reduction to a Mellin-separated form. Partition the range of u smoothly into dyadic blocks, with ramps of relative width $H^{-\kappa}$ at the two endpoints; the transition zones contain $\ll H^{0.53-\kappa}$ values of u and contribute $\ll H^{1-\kappa}(\log H)^C$ by $\mathcal{P}_u(Y) \ll Y$. Fix a block, $U \in [H^{0.47}/4, 2H^{0.53}]$, $Y = H/U$, $a = \log U / \log H \in [0.46, 0.54]$. Then:

- (a) the dilation average contributes $\ll U(\log H)^C$, since $\sum_{d'} (\lambda(d')/d') |\bar{G}_{d'}(H/u)| \ll (\log Y)^C$ for every u ;
- (b) the moduli $d' > YH^\eta$ contribute $\ll H^{1-\eta+3\varepsilon'}$ by part III's Step 3, which is valid for every U ;
- (c) the moduli $d' \leq Y^{1-\delta}$ contribute $\ll UY^{1-\delta}$, by $|(\lambda/d')G^\circ| \leq |\lambda|\omega(1+\sigma/(8d'\varphi))$ and Shiu's theorem [7];
- (d) on the remaining range the finite Fourier expansion of part III applies at $N(u) = \lfloor H/u \rfloor$, the frequency $k = 0$ cancelling exactly because $R_{d'}$ is symmetric and $0 \notin R_{d'}$; the frequencies $k > K_1$ contribute $\ll H^{1-\eta}$, and replacing $\xi_k(d', N(u))$ by $\xi_k(d', H/u)$ costs $\ll U(\log H)^C$. Note $d' \geq Y^{1-\delta} > 2K_1$, so $(k, d') \leq K_1 < d'$;
- (e) with $F_{k,d'}(v) = \lambda(d')\xi_k(d', H/v)\phi(v/U)\psi(v)/(2d'^2)$ one has, for $T_0 = 1 + kY/d' + H^\kappa$ and $J \in \{0, 1, 2\}$, $\int |\tilde{F}_{k,d'}(it)|(1+|t|)^J dt \ll |\lambda(d')|(Y/(kd'))T_0^J \log H$, by the total-variation bound for $|t| \leq T_0$ and $J+2$ integrations by parts beyond it, using that the ramp's derivatives are supported on a set of relative measure $\ll H^{-\kappa}$; summing, $\sum_{k \leq K_1} \sum_{d'} |\lambda|\omega \int |\tilde{F}_{k,d'}| \ll Y(\log H)^3$.

The block therefore equals $(2\pi)^{-1} \int \sum_{k \leq K_1} \sum_{d'} \tilde{F}_{k,d'}(it) \mathcal{S}_{k,d'}(t) dt$ plus the errors above, where $\mathcal{S}_{k,d'}(t) = \sum_u w(u)u^{-it} \mathbf{1}_{(u,d')=1} [\rho_k(d'; u^*) - \omega(d')\epsilon_{d'}(k)]$, and u^{-it} factors over any factorisation of u . The mean terms contribute $\ll UY^\delta(\log H)^C$.

Step R5: decomposition and covering. Write $w = c_f(1 * \chi_D * \gamma)$ with χ_D the primitive character of the field, γ multiplicative with $\gamma(p) \ll 1/p$, $|\gamma(n)| \leq 20^{\omega(n)}$ and $\sum_n |\gamma(n)| n^{-1/2-\varepsilon} < \infty$ (for $\omega(2) = 1$ put $u = 2u'$ and absorb the Euler factor at 2 into γ). The part with $n_3 > H^{\delta_3}$ is $\ll UYH^{-\delta_3/2+\varepsilon}$. Decompose n_1, n_2, n_3 into smooth dyadic boxes N_1, N_2, N_3 , discarding boxes with $N_1 N_2 N_3 \not\equiv U$, which vanish. For $a \in [0.46, 0.54]$ the boxes split into three cases: either some atom has $N_i \geq Y^{1/2} H^{3\delta}$ (Step R2); or some atom lies in $W = [H^{3\delta}, UY^{-1/2} H^{-3\delta}]$ (Step R3); or both $N_1, N_2 \in [UY^{-1/2} H^{-3\delta}, Y^{1/2} H^{3\delta}]$ (Step R4). Indeed if $N_2 < H^{3\delta}$ then $N_1 \gg UH^{-3\delta-\delta_3} \geq Y^{1/2} H^{3\delta}$, because $U/Y^{1/2} \geq H^{0.19}$ on the block; the covering is exhaustive with margin 0.183.

Step R2 (Type I). Let an atom of length $N \geq Y^{1/2} H^{3\delta}$ carry the weight 1 (or χ_D restricted to a class modulo the conductor), the cofactor m' carrying arbitrary coefficients $\ll \tau_3(m') 20^\omega$. For fixed k, d', r, m', t , with $g = (k, d')$ and $m = d'/g$, the atom sum is $\sum_n W(n) e_m((k/g)r(nm')^*)$ with $W(n) = \phi(n/N) n^{-it}$ after removing $(n, g) = 1$ by Möbius, so Lemma 1 with $T = 1 + |t|$ applies and gives $\ll H^\varepsilon (1 + |t|)^2 (1 + N/m) m^{1/2}$. Summing over $m' \ll U/N$, the roots, the moduli against $\int |\tilde{F}|(1 + |t|)^2$ and $k \leq K_1$, and splitting the two branches of $T_0^2 \ll 1 + (kY/d')^2 + H^{2\kappa}$, the contribution is $\ll UY(H^{-3\delta+4\eta+1.5(1-a)\delta} + H^{-3\delta+\eta/2+2\kappa}) + UY^{1/2+O(\delta)} \ll UY H^{-1.79\delta}$.

Step R3 (Type II). Let $u = u_1 u_2$ with $|a(u_1)|, |b(u_2)| \ll H^\varepsilon$ and $U_2 \in W$. Apply Cauchy-Schwarz over (d', u_1) with weight $|\beta(d')|$, $\beta = \tilde{F}_{k,d'}(it)$, replacing the u_1 -range by a smooth majorant. The diagonal gives $S_1 \ll H^{2\varepsilon} U_1 \sum_{d'} |\beta|$. In S_2 the terms carrying $\epsilon_{d'}$ are negligible, and the main u_1 -sums are $\sum V(u_1/U_1) e_{d'}(\nu u_1^*)$ with $\nu \equiv k(ru_2^* - r'u_2'^*)$; Lemma 1 bounds each by $\tau(d')(1 + U_1/d') d'^{1/2} (\nu, d')^{1/2}$, and trivially by $3U_1$. Writing $(\nu, d') = (k, d')e$ with $e = \prod_{p|d', p \nmid k, ru_2' \equiv r'u_2(p)} p$, for fixed (d', r, r', u_2) there are at most $4U_2/e$ values of u_2' with that e when $e \leq 2U_2$, and at most one when $e > 2U_2$ — the latter case containing the diagonal and all degenerate pairs, since $d'/(k, d') > 2U_2$. Hence $S_2 \ll H^{O(\varepsilon+\eta)} \sum_{d'} |\beta| [U_2^2 Y^{1/2} + U_1 U_2^2 Y^{-1/2} + U_1 U_2]$, the middle term alone carrying the loss $Y^{\delta/2}$, and therefore the contribution is $\ll UY H^{-1.45\delta}$.

Step R4 (the residual). Here both atoms lie in $[UY^{-1/2} H^{-3\delta}, Y^{1/2} H^{3\delta}]$. For fixed k, d', r, n_3, t , with $g = (k, d')$ and $m = d'/g$ squarefree, $m \geq Y^{1-\delta} H^{-2\eta}$, the numerator $(k/g) r n_3^*$ is a unit modulo m and the double sum is of the shape treated in Corollary 4, with the coprimality conditions and the dyadic cut-offs absorbed into the coefficients. On the block, $\log N_i / \log m \geq 0.3462 > 1/3$ and $2N_i \leq m$, so Corollary 4 applies and gives $\ll m^\varepsilon (N_1 N_2)^{15/16}$. Since $N_1 N_2 \asymp U/n_3$, summing over n_3 against $|\gamma|$, over the roots, over the moduli against $\int |\tilde{F}|$ and over $k \leq K_1$ gives $\ll UY H^{-0.028}$.

Conclusion. Per block the errors are $H^{1-4\eta/7}$ from R1(b), $H^{1-\eta}$ from R1(d), $UY^{1-\delta}$ from R1(c), UY^δ from the mean terms, $H^{1-\delta_3/2+\varepsilon}$ from the n_3 -tail, $H^{1-\kappa}(\log H)^C$ from the ramps, and the three savings above. With $O(\log H)$ blocks the binding constraint is R1(b), and Theorem 5 follows for every $\delta_M < 4\eta/7$. $\square$

5 The small dilations

Throughout this section $D < 0$. We write h for a positive integer, $\Gamma_0(q)$ for the Hecke congruence subgroup, $\Gamma_0^0(u) = \{\gamma : u \mid b, u \mid c\}$, and for positive definite forms $g_i = \begin{pmatrix} A_i & B_i \\ B_i & C_i \end{pmatrix}$ we put $\langle g_1, g_2 \rangle = A_1 C_2 + A_2 C_1 - 2B_1 B_2$, so that the point-pair invariant of the associated points of the upper half plane is $u(g_1, g_2) = (\cosh d - 1)/2$ with $\cosh d = \langle g_1, g_2 \rangle / (2\sqrt{\det g_1 \det g_2})$. The kernel method of Grimmelt and Merikoski is used in the form of their Theorem 8.1 [4] and

Lemma 3.1 [3]; we follow their notation, in particular Δ_q , K_q , the kernel quantities $\mathcal{K}_1(q; Z, R)$, $\mathcal{K}_2(q; Z)$ and the functional α_q on the Heegner set.

5.1 The family of a dilated piece, and its normal form

The piece $\mathcal{P}_u(Y)$ counts the roots of $Q_u(\ell) = u^2\ell^2 + |D|$ modulo admissible k , which are coprime to $2Du$. The content of Q_u is $c = \gcd(u^2, |D|)$, and since u is squarefree, $v_p(c) = \min(2, v_p(D))$ for $p \mid u$, so $c = Ps^2$ with

$$P = \prod_{p \mid u, v_p(D)=1} p, \quad s = \prod_{p \mid u, v_p(D) \geq 2} p, \quad u' = u/(Ps), \quad a = Pu'^2, \quad h = \frac{|D|}{Ps^2}, \quad h_0 = Ph = \frac{|D|}{s^2}.$$

Then $\gcd(a, h) = 1$, $P \parallel h_0$, $h_0 = O_D(1)$, and $(u', h_0) = 1$; moreover u' is odd: either D is odd and w is supported on odd u (a root of f modulo 2 would be a fixed prime divisor), or D is even, hence $4 \mid D$ (discriminants are $\equiv 0, 1 \pmod{4}$), and the prime 2 enters s . Since the primes of c divide u and k is coprime to u , $k \mid Q_u(\ell)$ if and only if $k \mid a\ell^2 + h$: the piece is a Type I sum for $a\ell^2 + h$ in the sense of [3], and by Lemma 3.1 there the pairs (k, ℓ) with $k \equiv 0 \pmod{q_0}$ correspond to the family

$$\mathcal{S}_{a,h}(q_0) = \left\{ g = \begin{pmatrix} A & B \\ B & C \end{pmatrix} : AC - B^2 = ah, A, C > 0, a \mid B, aq_0 \mid C \right\}$$

at level $q = aq_0$, the functional being $\alpha_q(g) = \mathbf{1}_{q \mid c(g)} \mathbf{1}_{a \mid b(g)}$ on the Heegner set Λ_{ah} (their $b^2 - ac = ah$ is a slip for $ac - b^2 = ah$). The case $P = s = 1$ is $a = u^2$, the case treated in our earlier drafts; the general normal form costs nothing extra, as Lemma 6 shows.

5.2 The off-diagonal at one level does not grow with the dilation

Lemma 6. *Let $(a, h) = (Pu'^2, |D|/(Ps^2))$ be as in §5.1, $F = \mathcal{S}_{a,h}(1)$, and for $Z \geq 1$*

$$\text{off}(a; Z) = \sum_{g_2 \in \Gamma_0(a) \setminus F} \sum_{\substack{g_1 \in F, g_1 \neq g_2 \\ u(g_1, g_2) \leq Z}} (1 + u(g_1, g_2))^{-1/2},$$

the orbits counted with the $\text{SL}_2(\mathbb{Z})$ -stabiliser weights $|\Gamma_g|^{-1} \leq 1$. Then $\text{off}(a; Z) \ll_{D, \varepsilon} (Zu)^\varepsilon (1 + Z^{1/2})$, uniformly in u : the u -dependence is the divisor factor $12^{\omega(u')}$, absorbed by $(Zu)^\varepsilon$, and there is no power of h .

Proof. (1) *Undilation.* Let $\delta = \text{diag}(1, 1/u')$ and $G = \delta g \delta$. For $g \in F$ write $B = au'^2 B_1 \dots$ more precisely $B = aB_1$, $C = aC_1$; then $G = \begin{pmatrix} A & Pu' B_1 \\ Pu' B_1 & PC_1 \end{pmatrix}$ has $\det G = ah/u'^2 = h_0$, and $g \mapsto G$ is a bijection of F onto $F' = \{G : \det G = h_0, Pu' \mid B_G, P \mid C_G\}$, with $\delta \Gamma_0(a) \delta^{-1} = \{\gamma : u' \mid b, Pu' \mid c\} = \Gamma_0(P) \cap \Gamma_0^0(u')$ (using $(P, u') = 1$). The map is an isometry of the associated points, and $\langle g_1, g_2 \rangle = u'^2 \langle G_1, G_2 \rangle =: u'^2 N$ with $1 + 2u(g_1, g_2) = N/(2h_0)$. Hence $N \in \mathbb{Z}$, $N > 2h_0$ for $g_1 \neq g_2$, and

$$\text{off}(a; Z) = \sum_{2h_0 < N \leq 2h_0(1+2Z)} \left(\frac{4h_0}{N + 2h_0} \right)^{1/2} P_u(N), \quad P_u(N) = \#\{(\Gamma_0(P) \cap \Gamma_0^0(u'))\text{-orbits of pairs } (G_1, G_2) \in F\}$$

(2) *Lift to level one.* The stabiliser in $\text{SL}_2(\mathbb{Z})$ of a pair of distinct points is $\pm I$, so each orbit lies in a unique $\text{SL}_2(\mathbb{Z})$ -orbit of pairs of determinant- h_0 forms with $\langle G_1, G_2 \rangle = N$, and $P_u(N) = \sum_{\mathcal{O}} \#\{\tau \in (\Gamma_0(P) \cap \Gamma_0^0(u')) \setminus \text{SL}_2(\mathbb{Z}) : \tau.G_1, \tau.G_2 \in F'\}$ over the $P_1(N)$ such orbits, evaluated at any representative. Taking G_2 among the reduced forms of determinant h_0 and observing that $W = 2h_0 G_1 - N G_2$ lies in the rank-two lattice $G_2^\perp$ with $-\det W = h_0(N^2 - 4h_0^2)$, the

number of representations of a fixed integer by a definite binary lattice gives $P_1(N) \ll_D N^{o(1)}$ — here $h_0 = O_D(1)$ is what makes the bound uniform. (3) *The local count.* By the Chinese remainder theorem the coset space factors over the primes of Pu' ; parametrise $T_p \backslash \mathrm{SL}_2(\mathbb{F}_p)$ by the ordered pair of distinct lines spanned by the rows of τ , and $\Gamma_0(p) \backslash \mathrm{SL}_2(\mathbb{F}_p)$ by the second row $L_2 \in \mathbb{P}^1(\mathbb{F}_p)$. Then $p \mid B(\tau.G)$ reads $L_1 \perp_G L_2$ and $p \mid C(\tau.G)$ reads $Q_G(L_2) = 0$. For $p \mid u'$ (so $p \nmid 2h_0$) the condition is $u' \mid B$ only: for one form there are $p - \chi_{-h_0}(p)$ cosets; for a pair, L_2 must be an eigenline of $G_2^{-1}G_1$, whose characteristic polynomial is $\lambda^2 - (N/h_0)\lambda + 1$, so there are at most 2 cosets unless $G_2^{-1}G_1$ is scalar modulo p , in which case $p^2 \mid N^2 - 4h_0^2$ and there are at most $p + 1$. For $p \mid P$ both conditions are imposed, and they collapse to one: $p \mid B_G$, $p \mid C_G$ and $p \parallel h_0 = \det G$ force $p \nmid A_G$, so $G \equiv \mathrm{diag}(A_G, 0)$ is of rank one modulo p , its unique projective zero is its kernel line L_0 , and for $L_2 = L_0$ one has $GL_2 \equiv 0$, so $L_1 \perp_G L_2$ for every L_1 . Hence the local factor at $p \mid P$ is 1, for one form and for pairs, and altogether the number of cosets for a pair is at most $2^{\omega(u')}g_2(N)$ with $g_2(N) = \prod_{p \mid u', p^2 \mid N^2 - 4h_0^2} p$. (4) *The sum over N .* With $M = 2h_0(1 + 2Z)$ and d running over the divisors of u' with $d \leq M$ (which is automatic, since $d^2 \mid N^2 - 4h_0^2 < M^2$),

$$\sum_{2h_0 < N \leq M} N^{-1/2} g_2(N) \leq \sum_{d \mid u', d \leq M} d \sum_{\substack{N \leq M \\ d^2 \mid N^2 - 4h_0^2}} N^{-1/2} \leq \sum_{d \mid u', d \leq M} d \cdot 2^{\omega(d)} (d^{-1/2} + 2M^{1/2}/d^2) \leq 3\tau_3(u')M^{1/2},$$

because the $2^{\omega(d)}$ classes $N \bmod d^2$ with $d^2 \mid N^2 - 4h_0^2$ each have $N \geq d$. Combining (1)–(4), $\mathrm{off}(a; Z) \ll_{D, \varepsilon} N^{o(1)} 2^{\omega(u')} \tau_3(u') M^{1/2} \ll (Zu)^\varepsilon (1 + Z^{1/2})$. $\square$

The mechanism is that the distances between family points are quantised: $\cosh d \in (2h_0)^{-1}\mathbb{Z}$ independently of the dilation, so the number of pairs at distance $\leq Z$ is governed by the divisors of $N^2 - 4h_0^2$, not by the size of the level. Numerically, for $a = 3u'^2$ and $a = 15u'^2$ ($D = -3, -15$) the diagonal is exactly $c \prod_{p \mid u'} (p - \chi_{-h_0}(p))$ with the factor at the primes of P equal to 1, and $\mathrm{off}(a; Z)/Z^{1/2}$ shows no growth in u' ; the local count of step (3) was also checked by enumerating $\mathrm{SL}_2(\mathbb{F}_p)$ for 141 normal forms, nine of them with D non-fundamental.

5.3 Theorem S

Theorem 7. *Let f be as above with $D < 0$. For every $c > 0$,*

$$\sum_{\substack{u \leq H^{1/3-c} \\ (u, D) = 1}} w(u) \mathcal{P}_u(H/u) \ll_{f, c} H^{1-\delta_S(c)}, \quad \delta_S(c) = \frac{2}{3} \cdot \frac{1}{9} \cdot \frac{3}{2} c(1 - 2\theta) \approx \frac{c}{12},$$

where $\theta = 7/64$ is the Kim–Sarnak bound. The parameters below may be taken as $\delta = \eta = \eta' = \frac{1}{9} \cdot \frac{3}{2} c(1 - 2\theta)$.

Remark 8 (on the coprimality restriction). Every step of the proof is written for the general normal form of §5.1; the only place where $(u, D) = 1$ is still used is the level sum of §5.3, whose restatement at (u', h_0) is recorded in the repository and awaits an independent reading. Once that is in, the restriction disappears. The prime 2 needs no hypothesis.

Step S1: the reduction

Fix u squarefree with $u \leq H^{1/3-c}$ and $Y = H/u$. Since $R_k^{(u)}$ is symmetric and $0 \notin R_k^{(u)}$ (as k is coprime to $2D$), pairing $x \leftrightarrow k - x$ in the definition of the piece gives the exact identity

$$\sum_{x \in R_k^{(u)}} B_k^{(Y)}(\langle x \rangle_k) = \frac{1}{2} \sum_{\ell \neq 0, k \mid a\ell^2 + h} T(\ell) - \frac{\varrho(k)}{2k} \int_{\mathbb{R}} T, \quad T(t) = (Y - |t|)^+, \quad \varrho(k) = |R_k^{(u)}|, \quad (1)$$

the shape “count minus density times mass” of [3]. The small moduli $k \leq Y^{1-\delta}$ contribute $\ll \sum_{k \leq Y^{1-\delta}} |\lambda(k)| \varrho(k) \ll Y^{1-\delta} (\log Y)^{O(1)}$, since for each pair $\{x, k-x\}$ the bracket $B_k^{(Y)}(m) + B_k^{(Y)}(k-m)$ lies in $[-k/4, k/4]$ (part I, proof of Theorem 5) and Shiu’s theorem bounds the sum; no smoothing is needed there, the identity being exact for each k . The large moduli $k > Y^{1+\eta}$ contribute $\ll Y^{1-\eta} (\log Y)^{11}$ by part III’s Lemma tail, uniformly in $u \leq Y^3$. To hand a smooth weight to the kernel theorem we insert ψ_1 , supported on $[Y^{1-\delta}/2, 2Y^{1+\eta}]$ and equal to 1 on $[Y^{1-\delta}, Y^{1+\eta}]$, and remove the two transition ranges by the same two bounds. On the window we write, in this order,

$$\lambda(k) \mu^2(k) \mathbf{1}_{(k, 2D)=1} = \sum_{j|(k, \text{rad } 2D)} \mu(j) \sum_{e|k} \kappa(e) \sum_{m^2|k} \mu(m),$$

with κ supported on split squarefree integers (so that $1 * \kappa$ reproduces λ only after the j -Möbius) and $(k, u) = 1$ automatic (a prime of (k, u) would divide $|D|$, hence $2D$); the tail $m > M := Y^\eta$ is removed *before* any further decomposition at a cost $\ll Y^{1-\eta+\varepsilon}$, and no truncation in e is needed. Then T is decomposed into smooth dyadic boxes $T_L(\ell) = Y \psi_2(\ell/L)$ in $|\ell| \sim L$, the two kinks of T at $\ell = \pm Y$ smoothed at scale $\Delta = Y^{1-\eta'}$, which changes the sum by $\ll Y^{1+\delta-2\eta'+\varepsilon}$ in total and gives $\psi_2^{(J)} \ll Y^{J\eta'}$. A box (L, K) , K dyadic in $[Y^{1-\delta}, Y^{1+\eta}]$, has trivial size $\ll (Y/K) L H^\varepsilon$, so the boxes with $L \leq Y^{1-3\delta}$ contribute $\ll Y^{1-2\delta+\varepsilon}$ together and the kink of T at $\ell = 0$ never matters. The outcome: for every u ,

$$\mathcal{P}_u(Y) = \sum_{\text{boxes } (L, K)} \sum_{j, e, m \leq M} \mu(j) \kappa(e) \mu(m) \mathfrak{B}(L, K; q_0) + O(Y^{1-\min(\delta, \eta, 2\eta'-\delta)+\varepsilon}), \quad q_0 = \text{lcm}(j, e, m^2), \quad (2)$$

with $O(\log^2 H)$ boxes, $Y^{1-3\delta} \leq L \leq Y$, and $\mathfrak{B}(L, K; q_0) = \frac{1}{2} \sum_{k \equiv 0 (q_0)} \frac{1}{k} \psi_1(k/K) [\sum_{k|a\ell^2+h} T_L(\ell) - \varrho(k) \int T_L/k]$; the levels are $q = a q_0$ with $\gcd(a, q_0) = 1$ (for $p \mid P$, $k \mid a\ell^2 + h \equiv h \pmod{p}$ and $p \nmid h$ force $p \nmid k$; for $p \mid u'$, $p \nmid 2D$ and $(k, u) = 1$).

Step S2: the two-scale form

Following §5 of [3], replace T_L by a second weight $T^{(2)}(\ell) = (\int T_L) \psi(\ell/X_2) / (X_2 \int \psi)$ of the same mass at the scale $X_2 = K^{1+\eta} > L$ (i.e. $\eta(1-\delta) > \delta$, which the parameters satisfy). The density terms of the two scales are identical, so $\mathfrak{B}(L, K; q_0) = \mathcal{A}_1 - \mathcal{A}_2 + \mathcal{E}$, where $\mathcal{A}_i = \langle I | \Delta_q F_i | \alpha_q \rangle$ is built from (ψ_1, T_L) , resp. $(\psi_1, T^{(2)})$, exactly as in their §5, and $\mathcal{E}$ is the second-scale sum $\sum_{k \equiv 0 (q_0)} \frac{1}{k} \psi_1(k/K) [\sum_{k|a\ell^2+h} T^{(2)}(\ell) - \varrho(k) \int T^{(2)}/k]$, which Poisson summation modulo k evaluates: for $k \sim K$ the range X_2 exceeds k by the factor K^η , the dual frequencies are $\geq K^\eta$, and $\mathcal{E} \ll_A K^{-A}$ for every A , uniformly in u (the constants depend on A, η, ψ). The skew of F_1 is $X/Y = Lu|D|^{-1/2}$ with x -range $R = L/K \in [Y^{-\eta-3\delta}, Y^\delta]$; that of F_2 is larger by X_2/L , and its bound is smaller than that of $\mathcal{A}_1$ by $(L/X_2)^{1/2-\theta}$, so only $\mathcal{A}_1$ is estimated below.

Step S3: the hypotheses of the kernel theorem

In Iwasawa coordinates $x = \ell/k$, $y = \sqrt{h}/(uk)$ the weight of $\mathcal{A}_1$ is $F(g) = (Y/K) f(x, y)$ with $f(x, y) = \psi_2(\ell/L) \tilde{\psi}_1(k/K)$, $\tilde{\psi}_1(t) = \psi_1(t)/t$; f is supported on $|x| \asymp L/K$, $y \asymp \sqrt{h}/(uK)$, is not a product (the argument of ψ_2 depends on y too), and each ∂_y produces a factor $1/y$, so $\partial_x^{J_1} \partial_y^{J_2} f \ll (Y^{\eta'})^{J_1+J_2} (L/K)^{-J_1} (\sqrt{h}/(uK))^{-J_2}$, the only non-dyadic scale being the kink smoothing. After splitting the x -support (of ratio ≤ 4) into $O(1)$ smooth dyadic pieces, $f \in C_{\delta_{\text{GM}}}^{10}(X, Y)$ in the sense of Definition 2 of [4] with $\delta_{\text{GM}}^{-1} \asymp Y^{\eta'}$, and $X/Y = Lu|D|^{-1/2} > \delta_{\text{GM}}$ on every box. The Hecke variable is trivial, so their Rankin–Selberg input is not invoked; $q^{o(1)} = (u^2 q_0)^{o(1)} = H^{o(1)}$; and the unspecified $\delta_{\text{GM}}^{-O(1)}$ of Theorem 8.1 is δ_{GM}^{-C} with $C = 6$ by Remark ???. Theorem 8.1 then

gives, for every $Z_0, Z_1, Z_2 \geq 1$ with $Z_0 Z_1 Z_2 \geq X/Y + 1$,

$$|\mathcal{A}_1| \ll H^{o(1)} Y^{C\eta'} \frac{Y}{K} (Lu)^{1/2} Z_0^\theta \mathcal{K}_1(q; Z_1, R)^{1/2} \mathcal{K}_2(q; Z_2)^{1/2}. \quad (3)$$

Step S4: the first kernel quantity

The example in §1.2 of [4] gives, for $q \geq 1$, $Z \geq 1$ and $0 < R \leq H^{O(1)}$, $\mathcal{K}_1(q; Z, R) \ll H^{o(1)}(1 + R + 1/(qR) + Z/q)$ (the cases $c = 0$, $b = 0$, $bc \neq 0$ of $\gamma \in \Gamma_0(q)$; we checked it numerically at our levels). Summed over the levels $q = a \operatorname{lcm}(j, e, m^2)$ of a block $e \sim E$, $m \sim M$, with $1/R \leq Y^{\eta+3\delta}$ and $R \leq Y^\delta$,

$$\sum_{\text{levels}} \mathcal{K}_1 \ll_D H^{o(1)} Y^{3\delta+\eta} (EM + Z_1/a). \quad (4)$$

Step S5: the second kernel quantity, the dictionary and the diagonal

By definition (§4.1 of [3]) and since $\alpha_q \geq 0$ and the subtracted volume term is ≥ 0 , $\langle \alpha_q | \Delta_q k | \alpha_q \rangle \leq \langle \alpha_q | K_q k | \alpha_q \rangle$, and the smooth kernel may be replaced by the majorant $\mathbf{1}_{u \leq Z_2^2} (1+u)^{-1/2}$. Unfolding the γ -sum once gives the orbit form $\operatorname{diag}(q) + \operatorname{off}(q; Z_2)$ with the $\operatorname{SL}_2(\mathbb{Z})$ -stabiliser weights on the g_2 -side and the g_1 -side unweighted (with $\Gamma_0(q)$ -stabilisers the identity fails by a factor 2). Unfolding a second time, as in Proposition 4.1 of [3],

$$\operatorname{off}(aq_0; Z_2) = \sum_{z_2 \in \Lambda_{ah}} |\Gamma_{z_2}|^{-1} \sum_{\substack{w_1 \in \mathcal{S}_{ah} \\ w_1 \neq z_2}} k(u(w_1, z_2)) n_{aq_0}(w_1, z_2), \quad n_q(w_1, z_2) = \#\{\tau \in \Gamma_0(q) \backslash \operatorname{SL}_2(\mathbb{Z}) : \alpha_q(\tau \diamond w_1) = z_2\} \quad (5)$$

with $\mathcal{S}_{ah}$ the set of all forms of determinant ah , and for $\gcd(a, q_0) = 1$ the Chinese remainder theorem gives $n_{aq_0} = n_a n_{q_0}$ with $n_{q_0} = \#\{(c_0 : d_0) \in \mathbb{P}^1(\mathbb{Z}/q_0) : q_0 \mid Q_{w_1}(c_0, d_0), q_0 \mid Q_{z_2}(c_0, d_0)\}$. Summing n_a over the $\operatorname{SL}_2(\mathbb{Z})$ -orbits of pairs with invariant N is the count $P_u(N)$ of Lemma 6: the a -part of the off-diagonal is that lemma, and the q_0 -part is the level sum below. The diagonal is the weighted number of $\Gamma_0(q)$ -orbits of the family, $\operatorname{diag}(q) = c_d(u) \rho^*(q_0)$ with $c_d(u) \asymp u$ exactly $\prod_{p|u'} (p - \chi_{-h_0}(p))$ times a factor $O_D(1)$, and $\rho^*(q_0) \leq 2^{\omega(q_0)} \prod_{p|(q_0, 2D)} (p+1) \ll_D 2^{\omega(q_0)}$ the number of projective zeros modulo q_0 ; hence $\sum_{\text{levels}} \operatorname{diag} \ll_D H^{o(1)} u EM$.

Step S6: the level-summed off-diagonal

Let $a = u^2$. For $1 \leq Z_2 \leq H^{O(1)}$ we claim

$$\sum_{q_0 \leq EM^2, (q_0, u)=1} \operatorname{off}(aq_0; Z_2) \ll_{f, \varepsilon} H^\varepsilon (1 + Z_2), \quad (6)$$

uniformly in u . Fix a pair (w_1, z_2) of distinct family points with invariant $N = \langle w_1, z_2 \rangle / u^2 \in (2h, M]$, $M = 2h(1 + 2Z_2^2)$. A common projective zero of Q_{w_1}, Q_{z_2} modulo p^k forces $p^k \mid \operatorname{Res} = \langle w_1, z_2 \rangle^2 - 4 \det w_1 \det z_2 = u^4(N^2 - 4h^2)$, because $x^3 \operatorname{Res}, y^3 \operatorname{Res} \in (Q_{w_1}, Q_{z_2})$, and Hensel's lemma gives at most 2 zeros of each form modulo p^k when $p \nmid 2ah$. The $O_D(1)$ primes $p \mid 2D$ are pulled out first with the trivial factor $p+1$ each, at most once since $v_p(q_0) \leq 1$ there (j is squarefree and $(em, 2D) = 1$); the primes of u do not divide q_0 , the family $\mathcal{S}_{u^2, h}(q_0)$ being empty otherwise ($Aq_0 C' - u^2 B'^2 = h$ would force $p \mid h$); so the u^4 is inert, and, the number of (j, e, m) with a given lcm being at most $\tau(q_0)^2 = H^{o(1)}$,

$$\sum_{(j, e, m): e \leq E, m \leq M} n_{\operatorname{lcm}(j, e, m^2)}(w_1, z_2) \leq \prod_{p|2D} (p+1) \cdot H^{o(1)} \sum_{q'_0 \mid N^2 - 4h^2, (q'_0, 2Dah)=1} 2^{\omega(q'_0)} \ll_D H^{o(1)} \tau_3(N^2 - 4h^2),$$

using $\sum_{d|n} 2^{\omega(d)} \leq \tau_3(n)$. The level sum therefore costs a divisor factor, not a factor EM^2 . Now $N^2 - 4h^2 \geq 4h + 1 > 0$ for distinct points, $\tau_3(N^2 - 4h^2) \ll M^\varepsilon$, $M \asymp hZ_2^2 \leq H^{O(1)}$, and Lemma 6 (steps (2)–(4), with $P_1(N) \ll (hN)^{o(1)}h^{1/2}$ and $\sum_{2h < N \leq M} N^{-1/2}g_2(N) \leq 3\tau_3(u)M^{1/2}$) give $\sum_{q_0} \text{off}(aq_0; Z_2) \ll H^\varepsilon 2^{\omega(u)}\tau_3(u)h^{3/2}M^{1/2} \ll H^\varepsilon(1+Z_2)$, the u -factor $6^{\omega(u)}$ being pointwise a divisor-size factor and averaged in the application by $\sum_{u \sim U} 6^{\omega(u)}/u \asymp (\log U)^5$. This proves (6). (The hypothesis $Z_2 \leq H^{O(1)}$ is where the ε -powers, all of which are M^ε , become H^ε ; it holds in Step S7 after the dichotomy there.) Together with Step S5, $\sum_{\text{levels}} \mathcal{K}_2 \ll H^{o(1)}(uEM + Z_2)$.

Step S7: the per-box bound

Fix a box (L, K) and a block (E, M) of levels, and apply (3) to every level with the same Z_0, Z_1, Z_2 . Pull the coefficients out by $|\kappa(e)\mu(m)| \ll H^{o(1)}/E$ — μ has no decay in m , and this costs a factor M — and apply Cauchy–Schwarz over the EM levels:

$$\sum_{\text{levels}} |\kappa(e)\mu(m)| |\mathcal{A}_1| \ll H^{o(1)} Y^{C\eta'} \frac{Y}{K} \frac{1}{E} (Lu)^{1/2} Z_0^\theta \left(\sum \mathcal{K}_1 \right)^{1/2} \left(\sum \mathcal{K}_2 \right)^{1/2}.$$

If $uEM \leq L^{1/2}$ choose $Z_1 = u^2EM$, $Z_2 = uEM$ and $Z_0 = Lu/(Z_1Z_2) = L/(u^2E^2M^2) \geq 1$; then by (4) and Step S6 the right-hand side is $\ll H^{o(1)} Y^{C\eta'+3\delta+\eta} \frac{Y}{K} \frac{L}{EM} \cdot M \left(\frac{u^2E^2M^2}{L} \right)^{1/2-\theta}$: the block's own trivial size $(Y/K)L/(EM)$ times $M(u^2E^2M^2/L)^{1/2-\theta}$. If $uEM > L^{1/2}$ the block is bounded by its trivial size $(Y/K)L/(EM) < (Y/K)L^{1/2}u$, and these blocks sum geometrically. Since the block bound decays like $E^{-2\theta}$ against a trivial size like $1/E$, all dyadic E sum without truncation, and $M \leq Y^\eta$ gives $\sum_M M^{1-2\theta} \ll Y^{\eta(1-2\theta)}$. Hence, for the box,

$$\sum_{\text{blocks}} \ll H^{o(1)} Y^{C\eta'+3\delta+\eta} \frac{Y}{K} \left(L^{1/2+\theta} u^{1-2\theta} Y^{\eta(1-2\theta)} + L^{1/2} u \right) = \frac{Y}{K} L \cdot H^{o(1)} Y^{O(\delta+\eta+\eta')} \left[\left(\frac{u^2}{L} \right)^{(1-2\theta)/2} + \frac{u}{L^{1/2}} \right],$$

a relative saving $(u^2/L)^{(1-2\theta)/2} Y^{O(\delta+\eta+\eta')}$ against the box's trivial size, with $u^2/L \leq u^2/Y^{1-3\delta}$.

Step S8: parameters and summation

Summing over the $O(\log^2 H)$ boxes and adding the errors of (2), each piece satisfies

$$\mathcal{P}_u(H/u) \ll \frac{H}{u} \left[\left(\frac{u^3}{H} \right)^{(1-2\theta)/2} H^{C\eta'+2\eta+O(\delta)} + H^{-\min(\delta, \eta, 2\eta'-\delta)+\varepsilon} \right],$$

using $u^2/Y = u^3/H$. With $\theta = 7/64$, $\frac{3}{2}(1-2\theta) = 1.171875$, and for $u \leq H^{1/3-c}$ the first bracket is $\leq H^{-1.171875c+C\eta'+2\eta+O(\delta)}$. Take $C = 6$ (Remark ??) and $\delta = \eta = \eta' = t$ with $9t = 1.171875c$, so that $C\eta' + 2\eta + O(\delta) \leq 8t$ and the first term is $\leq H^{-t}$, while $\min(\delta, \eta, 2\eta' - \delta) = t$. Finally $\sum_{u \leq H^{1/3-c}} |w(u)|(H/u)^{1-t} \ll H^{1-t} \sum_{u \leq H^{1/3}} |w(u)|u^{t-1} \ll H^{1-t+t/3} (\log H)^{O(1)} = H^{1-2t/3+o(1)}$, since $\sum_{u \leq U} |w(u)| \ll U$. This is Theorem 7 with $\delta_S(c) = 2t/3 = \frac{2}{3} \cdot \frac{1.171875c}{9} \approx 0.087c$. $\square$

Remark 9 (verification record). Theorem 5 was read three times by independent model readers (the third confirming the merged text) and Theorem 7's components in five rounds, each recorded with its corrections in the repository (`research/explore/PROOFS-uniform.md`, §28–§46); the local count of Lemma 6 and the identities of Steps S5–S6 were brute-forced on small cases. The identities of part I on which the pieces rest are formally verified in Lean 4 (part I, *Data and code*). No human expert has checked this paper.

6 What remains: the band at $u = H^{1/3}$

Theorems 5 and 7 leave a neighbourhood of $\alpha = 1/3$ uncovered, and the two mechanisms fail there for unrelated reasons. On the upper side the balanced Type II atoms sit at $d^{1/4}$, where the available energy input for reciprocals is exactly at its threshold: what would be needed is a bound for the six-fold energy $J_6(N)$ below N^4 at $N \asymp m^{1/4}$, which is not known even for prime moduli. On the lower side the kernel method's diagonal, the number of orbits of the family, is $\asymp u$, and at $u \asymp Y^{1/2}$ it meets the trivial bound.

Three remarks fix what such a band costs, and what closing it would buy.

Remark 10 (a band is cheap, and a failing method is free). Inside any range of α one still has part III's Theorem E piece by piece, $\mathcal{P}_u(H/u) \ll (H/u)(\log H)^{-c} \log \log H$ uniformly in $u \leq H^{1-\varepsilon}$. Hence a band $[H^{\alpha_0-\delta}, H^{\alpha_0+\delta}]$ on which nothing better is available costs $\ll_f \delta \cdot H(\log H)^{1-c} \log \log H$, not $\delta H \log H$: a mechanism that degenerates can never make the total worse than what is already proved, and any band whose width tends to 0 improves Theorem E by that factor.

Remark 11 (a point gap costs $O(H)$). Suppose the savings on the two sides vanish only at $\alpha = 1/3$, say like $H^{-c|\alpha-1/3|}$, with implied constants that are powers of $\log H$. Summing over dyadic blocks, $\sum_k H e^{-c|k \log 2 - \frac{1}{3} \log H|} = H(1 + 2/(e^{c \log 2} - 1)) = O(H/c)$: the blocks near the point contribute $O(H)$ in total. Since the main term $c_{\text{off}}(f)H$ comes from $u > H^{1/2+\varepsilon}$ and the remaining blocks are error terms, this would give $\text{Off}_f^*(H) = O(H)$, which is Hypothesis (E) of part I [5] and yields $\sum_{h \leq H} (1 - h/H)(S_f(h) - C(f)^2) = -\frac{1}{2}C(f) \log H + O_f(1)$: a bounded second-order term.

Remark 12 (what A_f needs). The constant of Conjecture 1 of part I requires $\text{Off}_f^*(H) = c_{\text{off}}(f)H + o(H)$, hence a saving factor $\gg \log H$ uniformly on $|\alpha - 1/3| \ll \log \log H / \log H$; a fixed positive power saving at $\alpha = 1/3$ itself suffices. A linearly vanishing saving, however small the rate, gives $O(H)$ but never $o(H)$.

Remark 13 (what is actually binding). An audit of every implied constant in the chain (research/explore/PROOFS-uniform.md §36) shows that no divisor-size loss constrains the argument: the divisor factors all occur under sums and average to powers of $\log H$, the kernel theorem's $\delta^{-O(1)}$ is δ^{-20} , its $(AD)^{o(1)}$ is $(\log AD)^{O(1)}$, and its $q^{o(1)}$ is 1 in the present application. What binds is the constant in Bourgain and Garaev's Hölder step, which grows as $k_1 \asymp 2/(9|\alpha - 1/3|) \rightarrow \infty$: the band on which the estimate is effective has width $\asymp (\log \log H / \log H)^{1/3}$. With the in-band fallback on part III's Theorem E this gives, once the map is written out, $\text{Off}_f^*(H) \ll H(\log H)^{0.63+o(1)}$ for $D < 0$, while everything outside the band sums to $O(H)$. The step from there to $\text{Off}_f^*(H) = O(H)$ therefore needs one thing only: an estimate that survives at the exceptional sizes $N \asymp m^{1/4}$ — an energy bound J_6 below N^4 for composite moduli, or a different tool there. A level-averaged spectral bound for the twist $e(cr\bar{d}/x)$ with $x \asymp Y^{1/2}$ would serve the same purpose; neither is in the literature.

References

- [1] Jean Bourgain and Moubariz Z. Garaev. Kloosterman sums in residue rings. *Acta Arith.*, 164(1):43–64, 2014.
- [2] Javier Cilleruelo and Moubariz Z. Garaev. Concentration of points on two and three dimensional modular hyperbolas and applications. *Geom. Funct. Anal.*, 21:892–904, 2011.
- [3] Lasse Grimmelt and Jori Merikoski. On the greatest prime factor and uniform equidistribution of quadratic polynomials, 2025. arXiv:2505.00493.

- [4] Lasse Grimmelt and Jori Merikoski. Weighted averages of $\mathrm{SL}_2(\mathbb{R})$ automorphic kernel, Part I: non-oscillatory functions, 2025. arXiv:2505.00489.
- [5] Jasper Reichardt. The pair singular series of a polynomial I, 2026. Preprint.
- [6] Jasper Reichardt. The pair singular series of a polynomial III: the off-diagonal in Cesàro form, 2026. Preprint.
- [7] P. Shiu. A Brun–Titchmarsh theorem for multiplicative functions. *J. Reine Angew. Math.*, 313:161–170, 1980.